# Fold a Paper Circuit! Origami Meets Electronics in a Cool New Way

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What if you could make a **circuit** just by folding paper? That's exactly what researchers in Hong Kong have done! They created a special paper-like material that can **conduct electricity** when you bend or cut it. The secret is a **liquid metal ink** sprayed onto the paper. At first, the ink doesn't **conduct electricity** because each tiny metal droplet is wrapped in an **insulating layer**. But when you fold or cut the paper, the pressure breaks that layer, letting the droplets connect and form a path for electricity. You can then attach **LEDs** or small motors with tape to make your paper creation light up or move. The team made a toolkit called LiqMetCraft with software that helps you design your project. They tested three styles: Chinese papercutting, **origami**, and 3D models. The cost? Only about $1.80 for a 10-by-10-centimeter piece. Now anyone can blend art and electronics in a snap!

## Word Treasure

**circuit** -- A complete loop that electricity can flow through.

**conduct electricity** -- To allow electric current to move through a material easily.

**liquid metal ink** -- A special ink made of tiny metal droplets that can flow and then harden to carry electricity.

**insulating layer** -- A thin coating that stops electricity from passing through.

**LEDs** -- Small, bright lights that use very little electricity (short for light-emitting diodes).

**origami** -- The Japanese art of folding paper into shapes like animals and flowers.

## Background you need

Hong Kong (a special region in China) is a center for technology and creative research.

Origami, the art of paper folding from Japan, has inspired many engineers to design foldable robots, solar panels, and now even paper circuits.

A normal electric circuit needs a complete loop of metal wire, but here the folding action creates the connection by squishing tiny metal drops together.

## Break it down

**WHO:** Researchers in Hong Kong.

**WHAT:** Invented a paper that becomes a working circuit when you fold or cut it, using liquid metal ink.

**WHERE:** Developed in a university lab in Hong Kong.

**WHY:** The ink has metal droplets wrapped in an insulating layer; folding breaks the layer so the droplets connect and let electricity flow.

## Why it matters

You might soon be able to make a greeting card that lights up or a paper toy that moves, mixing your art skills with real electronics.

## Test what you remember

**1. Where did the researchers create this special paper?**

- Ⓐ Tokyo
- Ⓑ Seoul
- Ⓒ Hong Kong
- Ⓓ New York

**2. What makes the paper conduct electricity when folded or cut?**

- Ⓐ A hidden wire inside
- Ⓑ Pressure breaks the insulating layer around metal droplets
- Ⓒ Glue melts and creates a path
- Ⓓ Sunlight charges the paper

**3. Which of these can you attach to make your paper creation light up?**

- Ⓐ A pencil
- Ⓑ An LED
- Ⓒ A paperclip
- Ⓓ A rubber band

**4. What is the name of the toolkit designed for this project?**

- Ⓐ FoldElectro
- Ⓑ PaperCircuit Pro
- Ⓒ LiqMetCraft
- Ⓓ OrigamiLED

**5. Which style was NOT tested by the team?**

- Ⓐ Chinese papercutting
- Ⓑ Origami
- Ⓒ 3D models
- Ⓓ Pop-up books

**6. About how much does a 10-by-10-centimeter piece of the paper cost?**

- Ⓐ $10
- Ⓑ $5
- Ⓒ $1.80
- Ⓓ $0.50

Answer key (try the quiz first!): 1: C 2: B 3: B 4: C 5: D 6: C

## Questions to think about

**1. Positive: This is an exciting, low-cost way for kids to learn about circuits and create art that glows and moves, making STEM subjects more fun.**

**2. Negative: The paper might tear easily, and the circuits could stop working if the folded edges get damaged, so it may not last very long.**

**3. Neutral: This idea blurs the line between craft and technology, showing how everyday materials can be smarter than they look.**

**MY THOUGHTS (at least 20 words):**